DE-3908

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Perform 2's complement substraction of $(149_{10} - 21_{10})$.

2. What is the use of complements? With an example explain 1's and 2's complement.

3. With a block diagram explain Multiplexer and bring out its truth table.

4. Give the circuit diagram and Truth Table for Full adder.

5. Give the circuit diagram for one-stage logic circuit.

6. Explain the Basic flip-flop circuit using NAND gate.

7. With an example explain sequential circuit.

8. List any three input devices with their advantages and disadvantages.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. (a) Convert Hexadecimal to Octal $(1E.C)_{16}$.

   (b) Convert Octal to Hexadecimal $(714.06)_8$.

10. Simplify the following expression into sum of products using Karnaugh map

    $F(A, B, C, D) = \sum (0, 1, 2, 5, 8, 9, 10)$

11. Explain the Quine – McKluskey method for minimization of Boolean function.

12. Explain binary adder with the circuit diagram.

13. With the neat sketch explain four bit shift register.

14. Describe BCD counter with a neat sketch.

15. What is ROM? Explain different types of ROM.

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DISTANCE EDUCATION

M.Sc. (SE) (5 years Integrated) DEGREE EXAMINATION, MAY 2016.

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Write a short notes on
   (a) Keywords
   (b) Identifiers
   (c) Tokens
   (d) Data types.

2. Explain about Implicit and Explicit type conversions.

3. Explain Recursion with example.

4. Write a program that would sort a list of names in alphabetical order.

5. Explain structures with example.

6. Write a short note about Primitive operations on circular lists.

7. Discuss about Trees and their Applications.

8. Write a program for Towers of Hanoi.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. How to manage input and output operations? Explain it with example.

10. Write a program to determine whether a given number is 'odd' or 'even' and print the message 'NUMBER IS EVEN' or ' NUMBER IS ODD'.

    (a) Without using else option

    (b) With else option.

11. Define array and its types. Write a program for matrix multiplication.

12. Discuss in detail about Pointers with example.

13. Explain about File Operations with example.

14. Define Stack. How to implement push and pop operations in Stack?

15. How to define Linked List as a Data Structure? Discuss about getnode and freenode operations.

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DE–3910

Sub. Code
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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) Construct the truth table for $(P \vee \neg Q)$. (4)

   (b) Write in symbolic form the statements :

   (i) The crop will be destroyed if there is a flood.

   (ii) If either Terry takes Calculus or Ken takes sociology , then Larry will take English. (4)

2. (a) If $A = \{2, 5, 6\}, B = \{3, 4, 2\}$, $C = \{1, 3, 4\}$, then find $A - B$ and $B - A$. Also show that $A - B \neq B - A$ and $A - C = A$. (4)

   (b) Give the power sets of the following :

   (i) $\{a, b, c\}$

   (ii) $\{1, \phi\}$. (4)

3. Show that the functions $f(x) = x^3$ and $g(x) = x^{\frac{1}{3}}$ for $x \in \mathbb{R}$ are inverses of one another.

4. Show that $f : \mathbb{R} - \{3\} \to \mathbb{R} - \{1\}$ given by $f(x) = \dfrac{x-2}{x-3}$ is a bijection and find its inverse.

5. If relations R and S are both reflexive, then show that $R \cup S$ and $R \cap S$ are also reflexive.

6. (a) Give an example of a semi group but not a monoid. (4)

   (b) Show that if every element in a group is its own inverse , then the group must be abelian. (4)

7. (a) Define graph and digraph, and give example to each. (4)

   (b) What is the maximum number of edges of an undirected simple graph with n vertices? (4)

8. (a) Define :

   (i) Complete graph

   (ii) Bipartiate graph. (4)

   (b) How many different directed tress are there with the nodes? (4)

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Show that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$. (8)

   (b) Write an equivalent formula for $P \wedge \left(Q \underset{\leftarrow}{\rightarrow} R \right)$ which contains neither the biconditional nor the conditional. (7)

10. (a) Let $\mathbb{Z}$ be the set of integers and let R be the relation defined by $R = \{\langle x, y\rangle / x \in \mathbb{Z} \wedge y \in \mathbb{Z} \wedge (x, y)$ is divisible by 3$\}$. Determine the equivalence classes generated by the element of $\mathbb{Z}$. (8)

(b) Show that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x - 3$ is a bijection and find its inverse. Also, compute $f^{-1} \circ f$ and $f \circ f^{-1}$. (7)

11. Let $f : X \to Y$ be a function and $A, B \subset X$. Show that $f(A \cup B) = f(A) \cup f(B)$ and $f(A \cap B) \subset f(A) \cap f(B)$.

12. For any three sets A, B, C , show that

(a) $A \Delta B = (A \cup B) - (A \cap B)$

(b) $A \cup B = A \cup C$ and $A \cap B = A \cap C \Rightarrow B = C$. (8+7)

13. (a) Prove that every subgroup of an abelian group is a normal subgroup. (8)

(b) If $f : G \to G'$ is a homomorphism, then prove that f is 1-1 if and only if $\ker f = \{e\}$. (7)

14. (a) Define a planar graph. Give an example. (8)

(b) Prove that any connected graph with n vertices and $n - 1$ edges is a tree. (7)

15. Prove that a given connected graph G is an Euler graph if and only if all vertices of G are of even degree.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Year Integrated) DEGREE EXAMINATION, MAY 2016.

C PROGRAMMING – LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give one question to each candidate by lot system.

1. (a) Write a C program to C program to Convert Numbers to Roman Numerals.

   (b) Write a C program to find a whether the given string is PALINDROME or not.

---------- Cut here ----------

2. (a) Write a C program to perform the following string operations using switch statement.

   (i) String copy

   (ii) Compare two strings, and

   (iii) Length of the string.

   (b) Write a C program to find the product of two matrices.

---------- Cut here ----------

3. (a) Write a program to generate Fibonacci series using recursion.

   (b) Write a C program to convert octal number to decimal number.

4. (a) Write a C program to find the biggest number among three numbers.

(b) Write a C program to find simple interest using function. (Hint : SI=PNR/100).

---------- Cut here ----------

5. (a) Write a C program to read the array values using pointer.

(b) Write a C program to check the given number is prime or not.

---------- Cut here ----------

6. (a) Write a C program to find the roots of a quadratic equation using function definition.

$ax^2 + bx + c = 0$.

(b) Write a program to display the odd numbers within 100.

---------- Cut here ----------

7. (a) Write a C program to arrange the given random numbers in ascending order.

(b) There are 50 students in a class. Their names and marks in four different subjects are given. If a student takes more than 40 marks in each subject, then he is declared "PASS".

Otherwise "FAIL". And also find their grade using the following conditions.

Grade D – If the Average is >= 90

Grade C – If the Average lies between 81 – 90

Grade B – If the Average lies between 71 – 80

Grade A – If the Average lies between 60 – 70.

Write a C program to do above using structure concept.

---------- Cut here ----------

8. (a) Write a recursive function to evaluate n! and write a C program to evaluate the nPr. [Hint : $n\Pr = n!/(n-r)!$].

(b) Write a C program to check the person is eligible to vote or not {Hint : If the age is >=18, the person is eligible to vote].

DE–9211 **17**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

DATA STRUCTURES LAB

Time : 3 Hours Maximum : 100 Marks

One question should be given to each candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Write a C Program to Insert Item into Unsorted Array.

   (b) Write a C Program to Search an Item using Linear Search.

---------- Cut here ----------

2. (a) Write a C Program to Sort an Array using INSERTION SORT.

   (b) Write a C Program to implement Stack using Linked list.

---------- Cut here ----------

3. (a) Write a C Program to Insert Item as Last Node using Linked List.

   (b) Write a C Program to Implement Queue using Array.

4. (a) Write a C Program to Delete any Specific Node using Linked List.

   (b) Write a C Program to Sort an Array using SELECION SORT.

---------- Cut here ----------

5. (a) Write a C Program to Search an Item in a Sorted Linked List.

   (b) Write a C Program to Implement Stack using Array.

---------- Cut here ----------

6. (a) Write a C Program to Search an Item using BINARY SEARCH.

   (b) Write a C Program to Implement Queue using Array.

---------- Cut here ----------

7. (a) Write a C Program to Sort an Array using MERGE SORT.

   (b) Write a C Program to Implement Queue using Linked List.

---------- Cut here ----------

8. (a) Write a C Program to Search an Item using Linear Search.

   (b) Write a C Program to Insert item After a Specific Node using Linked list.

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Discuss principles of OOP.

2. Explain the purpose of function with example.

3. Discuss about the access specifier with example.

4. Explain the worning principles of pointers with example.

5. Explain copy constructor with example.

6. Describe the Polymorphism.

7. Illustrate any two types of Inheritance.

8. Write short notes on Pure Virtual Function with example.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write brief notes on benefits and principles of OOPS.

10. Explain with examples :
    (a) Constructor and Destructor
    (b) Inline function.

11. Discuss about issues with pointer reference with example.

12. Differentiate Function Over loading and Operator Over Loading.

13. Explain in detail about static and dynamic binding with example.

14. Explain briefly with suitable example :

    (a) Base class

    (b) Derived class

    (c) Abstract class.

15. Write brief notes on overloading binary operators with examples.

DE–3912

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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. What are the services provided by the operating system?

2. Explain in brief about semaphores.

3. Write a note on round robin scheduling.

4. Explain the function of interrupt handlers.

5. Give a note on deadlock prevention and avoidance.

6. Write a note on swapping.

7. Explain about virtual memory.

8. Give a note on directory structure.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the following :

   (a) Types of operating system (8)

   (b) System calls. (7)

10. Describe the following :

    (a) Policy driven scheduling (8)

    (b) Two level scheduling. (7)

11. Explain the following :

    (a) I/O devices (8)

    (b) Device controllers. (7)

12. Elucidate the device-independent I/O software and user-space I/O software.

13. How to manage the memory with bitmaps and linked list? Discuss.

14. Describe page replacement algorithms in detail.

15. Explain in detail about file storage and file servers.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

RELATIONAL DATABASE MANAGEMENT SYSTEM

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. What is a Database Management System? Mention its features.

2. What is the difference between Schema, Subschema and Instances?

3. Write the advantages and disadvantages of using relational databases.

4. Write short notes on Entity and Referential Integrity.

5. Mention some of the Databases types.

6. Write the importance of DML statements used for managing data within schema objects.

7. What are the difference between DDL, DML and DCL commands?

8. Explain briefly about Embedded SQL and its features.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the following :

   (a) Meta Data

   (b) Schema and Subschema in Data Base Management System

   (c) Data Independence.

10. Discuss the following :

   (a) Semantic issues in Relation Model

   (b) Navigation and Efficiency in Relation Model

   (c) Write the purpose of Optical Storage.

11. Describe the Oracle Architecture with neat diagram.

12. Detail on the following relation database concepts.

13. Write and explain the commands with their syntax to do the following oracle queries

   (a) Date/Time related queries

   (b) Data dictionary queries.

14. Give details of the following :

   (a) Modifying Tables

   (b) Renaming Tables

   (c) Copying another Table

   (d) Dropping a Table.

15. Explain briefly about Embedded SQL Commands and its purposes.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

C++ – LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a C++ program using operator overloading to find the different and total length of given two various tubes specifies in meters and centimetres.

   (b) Using pointers create a class and write a program to get names and display them in sorted order.

---------- Cut here ----------

2. (a) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

   (b) Write a program using pointers to get a given number of names and display them in ascending order.

---------- Cut here ----------

3. (a) Assume you want to generate a table of multiple of any number. Write a program that allows the user to enter the number and then generate the table, formatting it into ten columns and 20 lines.

   (b) Declare student name and marks (mark 1, marks 2, and mark 3) as private variables in a class. Create a function, named get ‘data ( ) to get input and create another function, named result ( ), to print the results. Students shall obtain a minimum of 50 marks to pass. Other wise they fail.

4. (a) Write a program to process students mark with the help of classes. All subjects mark must be greater 50 for pass otherwise fail.

(b) Create a class employee that contains an employee number employee name and address. Write a menu driven C++ program to get the 'n' number of employee details and display all details in employee name wise sorted order.

---------- Cut here ----------

5. (a) Write a program to add 2 complex numbers using operator overloading.

(b) Write a C++ program to Count the number of alphabets, number, special characters, spaces and words in a given line of text.

---------- Cut here ----------

6. (a) Write a program which uses an array of pointers to objects and sort then in the alphabetical sequence.

(b) Develop a program to illustrate a copy constructor so that a string may be duplicated into another variable either by assignment or copying.

---------- Cut here ----------

7. (a) Write a C++ program using dynamic constructors to concatenate two strings.

(b) Write a C++ program to find the volume of physical structure (cube, cylinder, rectangular box) using function overloading.

8. (a) Write a menu driven C++ program using operator overloading to add and subtract two given matrices. Define the matrices in a class.

(b) Write a program in C++ to display the elder among 3 persons using 'this' pointer.

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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

RELATIONAL DATABASE MANAGEMENT SYSTEMS – LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate from lot system.

1. (a) Create a table as 'host_blood_donors' with the following details :

(Donor number Number (8),

Donor name Character (15),

Donor age Number (2),

Sex Character(5),

Donor address Character(15)

Blood group Character(3))

Perform the following :

(i) Enter at least five records

(ii) List the Donor number in Ascending Order

(iii) List the Blood groups — A, A+ and B+ using comparison operator IN( ).

(iv) List the donors having 's' as the second letter.

(b) Create a table 'New_paper_vendor' with the following data :

(Week Number(3),

Bought Number(4),

Sold Number(4)

Return Number(4),

Profit Number(7,2),

Loss Number(7,2)

Perform the following :

(i) Enter at least nine records (Insert data only - Week, Bought, Sold and Return columns only)

(ii) Update the Field Profit and Loss.

(iii) List the week where the profit is high.

(iv) Find in which week the sale of newspaper is high.

---------- Cut here ----------

2. (a) Create a table Employee with the fields like empno, empname, job, sal, designation and dept.

(i) List the details of the employees whose name starts with 'A'.

(ii) List the employees who are getting salary more than 2500 (or) 7500 (Using comparison operator ANY)

(iii) List the details of the employees who are earning more than 3000 and less than 8000.

(b) Create a table as student with the following details

(studno, sname, mark1, mark2, mark3, tot, avg, grade)

(i) INSERT only the necessary columns.

(ii) UPDATE the total, average and grade columns.

(iii) Display the student details whose grade is 'A'.

(iv) List the name of the student whose names begin with 'S'.

---------- Cut here ----------

3. (a) Write a PL/SQL program to add 10 booster marks for all mark 1? Student (studno, sname, mark1, mark2, total, avg, result)

(b) Write a PL/SQL to give grade for all students

Student(studno, sname, mark1, mark2, total, avg, result, grade)

Condition: If avg less than 50 – Grade 'C'

If avg > = 50 and <60 – Grade 'B'

If avg > = 75 – Grade 'A'.

---------- Cut here ----------

4. (a) Create a table Employee with the fields like empno, empname, doj, job, sal, designation and dept.

(i) List the details of the employees who are working in 'MUMBAI'.

(ii) List the details of the employees who are earning more than all the 'MARKETING EXECUTIVES'

(iii) List the details of employees who are joined before 'Anand'.

(b) Create a table 'Employee' with the following data:

Empno, Ename, Job, Deptno, Deptname, Basicpay, DA, HRA, PF

Grosspay, Netpay.

Perform the following :

(i) Insert only the necessary column.

(ii) Compute yearly salary of any one employee.

(iii) Calculate the Net salary of all employees.

(iv) List the employees having 'a' as the second letter.

---------- Cut here ----------

5. (a) Create a table as 'Sales_order' with the following details :

(Deptno Number(4),
Item desc_Character(15),
Sales_price Number(7,2),
Qty Number(3),
Cost_price Number(7,2),
Profit Number(8,2))

Perform the following :

(i) Insert data - except profit column

(ii) List the department number in Descending order.

(iii) List the item where the profit is high.

(iv) List the item where the quantity on hand is more than 10.

(b) Create a table for Hospital Billing System with the following fields :

(Patient no, patient name, age, doctor name, patient type (IN/OUT))

(i) List the details of the IN type patients.

(ii) List the details of the patients whose age between 40 and 60.

(iii) List the maximum age of the patients by each type (IN/OUT) category.

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Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, MAY 2016.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the uses of Computer Networks?

2. Write short notes on ISDN.

3. Explain the sliding window protocols.

4. What is ALOHA? Explain.

5. Describe the concept of firewalls and its advantages.

6. What are the performance issues in transport layer? Discuss.

7. Write short notes on the following :

   (a) TCP

   (b) UDP.

8. Illustrate the fundamentals of WWW.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the OSI reference model with diagram.

10. Write a brief note on communication satellites.

11. Describe the error detection and correction codes.

12. Explain any two routing algorithms in detail.

13. Briefly explain the Internet control protocols.

14. Discuss the basic concepts of network security.

15. Explain the following :
    (a) Multimedia
    (b) MPEG standards.

DE-3915

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

INTERNET AND JAVA PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. How to connect Internet with your computer?

2. Define JAVA array and explain its types with example.

3. What is meant by exception Handling? Explain with example program.

4. Explain the basic structure of Applet Program with simple example.

5. What is HTML? Explain HTML Basic tags.

6. Discuss the uses of HTML frames with suitable example.

7. What are the uses of I/O streams? Write some I/O stream methods.

8. Discuss the following in Networking Basics :

   (a) Socket Overview

   (b) Internet Addressing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss on the following :

   (a) World Wide Web. (5)

   (b) Functions of Web Browser with example. (10)

10. What are the control statements available in JAVA? Explain with examples.

11. Explain the concept of Packages in JAVA with suitable example program.

12. Briefly discuss on the following :

   (a) AWT controls

   (b) JAVA servlets.

13. Explain various table tags in html with example program.

14. Write the control structures in java script with example program.

15. Explain in detail about JAVA BEANS and discuss the following :

   (a) BDK

   (b) Creating a JAR File.

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Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain the Linear Sequential Software Process model.

2. Brief on size oriented metrics.

3. Write short notes on reusable software resources.

4. Discuss on Error Tracking.

5. Explain change control in Software Configuration Management.

6. Write short notes on Software Prototyping.

7. Discuss on Software Design principles.

8. Explain the white box testing technique.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Describe the Evolutionary Software process models.

10. Detail on Software Project Estimation techniques.

11. Explain the following :
    (a) Earned Value Analysis
    (b) Tracking the schedule.

12. Describe the SCM process.

13. Detail on Requirements specification and validation.

14. Describe Data Design.

15. Explain the Basis path testing technique.

DE–3917

Sub. Code
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DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Describe the architecture of Raster Graphics system.

2. How a picture in a window is napped onto viewport? Explain.

3. Derive the perspective projection transformation matrix.

4. What do you mean by MPEG standard? Explain.

5. Give the most common features of audio editing software.

6. Explain how audio and video are used I multimedia.

7. Write note on animation techniques.

8. How is image data stored in multimedia database?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write the midpoint circle drawing algorithm and explain how it works with an example.

10. (a) How is B-spline curve different from Bezier Curve?

    (b) What do you mean by rational B-spline? How is it more useful than non-rational B-spline in drawing curves?

    (c) Describe rendering pipeline in OpenGL...

11. (a) What do you mean by perspective projection? How are vanishing points generated?

    (b) Write the transformation matrices for rotation about origin in 3D.

    (c) Explain the Cyrus-Beck algorithm for clipping a line with respect to a rectangular window.

12. Describe Phong shading model for rendering polygon surface. How is it better than Gourad shading model? How can you make Phong shading model fast?

13. (a) What do you mean by authoring tool? Describe the grouping of the authoring tools.

    (b) Define Hypermedia, Hypertext, Links, Anchors and Nodes.

14. What are the main design issues for the multimedia authoring system? Briefly explain each one of these issues.

15. (a) Explain where and why the magnetic storage is used in multimedia system?

    (b) Describe the operation of scanner. What criteria would you use for selecting scanner?

    (c) What are the main design issues for the multimedia authoring system? Briefly explain each one of these issues.

DE–3918

Sub. Code
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DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain any six UNIX command with examples.

2. Write short notes on :

   (a) sort

   (b) nl

   (c) uniq.

3. Write a UNIX program to print the given numbers in reverse order.

4. Explain about the unmask command.

5. Write about Sed with an example.

6. Explain about String functions.

7. Mention the korn shell features.

8. Discuss about the following :

   (a) Chown

   (b) mkdir

   (c) rmdir.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Describe the filter commands in detail.

   (b) Write short notes on :

      (i) fgrep

      (ii) awk.

10. (a) Discuss briefly about concatenating files.

    (b) Write short notes on command line editing.

11. (a) Explain briefly about the various mathematical functions available in Unix.

    (b) Write short notes on searching for file content.

12. Discuss about special parameters and changing positional parameters with examples.

13. (a) Explain briefly about the various mode file management in detail.

    (b) Write short notes on environmental variables.

14. Discuss about the following commands with suitable examples.

    (a) tr

    (b) cpio

    (c) eval

    (d) join

    (e) paste.

15. (a) How will you debugging scripts?

    (b) Write short notes on argument validation.

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DE–9214 **36**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

INTERNET AND JAVA — LAB

Time : Three Hours Maximum : 100 Marks

The Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write a Java script to create an Order Form to select the house articles.

   (b) Create a web page in the format of front page of a newspaper using Text link. Align the text with colors.

---------- Cut here ----------

2. (a) Write a VB script to create a calendar for given month and year.

   (b) Develop a single page advertisement for a shop to be opened shortly.

---------- Cut here ----------

3. (a) Write a Java program using Applet to display the dialogue menu in Applet.

   (b) Design and publish a web page for a college.

4. (a) Write a Swing program to create the Tabbed panels.

(b) Develop a web page for job recruitment agency in an IT industry.

---------- Cut here ----------

5. (a) Write a Java program using Applet to display the different colors and fonts.

(b) Develop a picture gallery having atleast 5 pages. Each of them is having several pictures with suitable information.

DE–9215

Sub. Code
37

DISTANCE EDUCATION

M.Sc. Software Engineering (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

UNIX AND SHELL PROGRAMMING — LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate from Lot system.

1. (a) Write a Menu Driven shell program to copy, edit, rename, and delete a file.

   (b) A Hospital maintains patient details. Write a shell program to list the following.

      (i) Patients in a blood group wise.

      (ii) Patients in age between 25 to 40.

      (iii) Exit.

---------- Cut here ----------

2. (a) Write a program for four-function calculator by using yacc command.

   (b) Write and execute a shell program to do the following:

      (i) Write given string at given row, column in BOLD, Normal and reverse video.

      (ii) Search a given string using awk with various options.

3. (a) Write and execute a shell program to generate leave status report of an organization.

(b) Write and execute a shell program to print monthly payslips for employees in an organization. Assume your own data and generate accordingly.

---------- Cut here ----------

4. (a) Write a shell program to solve $(1/1^2)+(2/2^2)+(3/3^2)+....+(10/10^2)$.

(b) Write a shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located. (using UNTIL statement).

---------- Cut here ----------

5. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it:

(i) File existence

(ii) File readable

(iii) File writable

(iv) Both readable and writable.

(b) Write a Menu Driven Shell program for the following:

(i) Today's date

(ii) Users of system

(iii) Quit to UNIX.

6. (a) Write a shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located. (using UNTIL statement).

(b) Write a shell program which accepts the name of a file from the standard input and then perform the following tests on it:

(i) Enter the 5 names in a file

(ii) Sort the name in existing file

(iii) List unsorted and sorted file

(iv) Quit.

---------- Cut here ----------

7. (a) Write a Menu Driven shell program to perform the following tasks

(i) Enter the sentence in a file

(ii) Search a given whole word in an existing file

(iii) Quit.

(b) Write a shell program to prepare electric bill for domestic consumers.

For first 100 units – Rs. 1.00/unit

For next 100 units – Rs. 1.50/unit

Above 200 units – Rs. 3.00/unit

Prepare the Electricity bill in the following format:

Customer no _____

Customer name _____

Current reading _____

Charge _____

8. (a) Write a shell program to read and write the content in a file.

(b) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "Pattern found" else display "Error message" Also check if right number of arguments are entered.

DE–3919

Sub. Code
41

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2016.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. How to create and using Controls in VB.

2. Differentiate between Object and Instances.

3. Write short notes on Displaying and printing Information in VB.

4. Explain the In-built functions.

5. Discuss about menus in VC++.

6. Describe Mouse and keyboard Events.

7. Explain Serialization.

8. Write brief Notes on OLE.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Brief notes on managing projects in VB with examples.

10. Explain Accessing database with data control with Example.

11. Explain about grid control with Example.

12. Write Brief notes on MFC File Handling with Example.

13. Discuss SDI and MDI Environment.

14. Brief Notes on dynamic Data transfer function and User interface Classes with Examples.

15. Discuss about Communication with Other Applications

DE–3920

Sub. Code
42

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

SOFTWARE PROJECT MANAGEMENT AND METRICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the pragmatic software cost estimation.

2. Briefly describe on achieving required quality.

3. Explain the principles of conventional software engineering.

4. Write brief notes on transitioning to an iterative process.

5. Discuss major mile stones and minor mile stones.

6. Explain briefly project organization.

7. Write a short note on Automation building blocks.

8. Describe briefly the seven core metrics.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the convention software managements.

10. Give a brief explanation about process of the artifacts.

11. Discuss transitioning to an iterative process. .

12. Write a brief explanation about various workflows of the process.

13. Briefly discuss on Iterative Process Planning,

14. Explain the management indicators and quality indicators.

15. Discuss briefly on :

    (a) Next generation software economics

    (b) Modern process transitions.

DE–3921

Sub. Code
43

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

WEB TECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. What are the advantages of XML over HTML?

2. What are the types of cascading style sheet? Explain with example

3. List out the key features of EJB.

4. Define servlets? What is the need for it?

5. What are the advantages of Java Beans?

6. What are the different types of system vulnerabilities?

7. Differentiate session vs cookies.

8. List out the objects used for database access using JDBC.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is DTD? Why do we use it with XML documents? Explain the different components of DTD with example.

10. Describe the life cycle of a java servlet.

11. Discuss the following :

    (a) Entity beans

    (b) Session beans.

12. Explain the steps in installing and testing Tomcat server.

13. Explain the various methods defined by servlet request and servlet response interfaces?

14. Give a detailed description of JSP debugging techniques.

15. Explain different types of JDBC drivers and discuss their advantages and disadvantages in brief.

DE–3922

Sub. Code
44

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2016.

DATA WAREHOUSING AND MINING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain any two kinds of data mining.

2. What is optimizing Data for mining?

3. Discuss fuzzy sets and logic.

4. Explain in details about pattern matching.

5. Describe the Neural network.

6. Explain the Clustering issues.

7. Illustrate any one Partitional algorithm.

8. Describe briefly on web mining applications.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the classification of data mining system.

10. Illustrate the MLE.

11. Explain any two summarization models.

12. Describe the clustering approaches.

13. Explain in detail about SingleLink and MST SingleLink algorithm for clustering.

14. (a) Explain BEA algorithm.

    (b) Explain-GA algorithm.

15. Briefly explain about Web Knowledge Mining Taxonomy.

DE–3923

Sub. Code
45

DISTANCE EDUCATION

M.Sc. (S.E.) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

DISTRIBUTED COMPUTING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on resource sharing.
2. Write short notes on the different types of networks.
3. Briefly discuss on group communication.
4. Explain briefly about OS layers.
5. Write short notes on digital signatures.
6. Write an overview on cryptography pragmatics.
7. Explain briefly about directory services.
8. What are nested transactions? Discuss.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly discuss on internet protocols.
10. What is Remote Method Invocation? Discuss.

11. Briefly discuss on Remote Procedure call.

12. Give an overview on cryptographic algorithms.

13. Give detailed notes on Sun Network File system.

14. Write detailed notes on atomic commit protocols.

15. Discuss briefly on distributed deadlocks.

DE–9216

Sub. Code
46

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

VISUAL BASIC AND VC++ – LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Using AX DLL or EXE, add a class module that would perform the following function :

   (i) Test whether the given number is perfect or not

   (ii) Armstrong or not

   (iii) Factorial.

   (b) Write a VC++ program to fill background of the client area with a bitmap.

---------- Cut here ----------

2. (a) Write a VC++ program to create a list box in window.

   (b) Using SDK, type program code for keyboard and mouse events and GDI objects.

3. (a) Create an employee with Empno, Name, Bpay, HRA, DA, PF, LIC, OP and NP with the following calculation :

HRA — 10% of Bpay

DA — 5% of Bpay

PF — 3% of Bpay

LIC — 5% of Bpay

GP — Bpay + DA + HRA

NP — GP — (PF + LIC)

Using RDO, implement the following operations.

(i) Insertion

(ii) Search and delete

(iii) Modify

(iv) Display.

(b) Write a VC++ program to find out whether a mouse is attached or not; and if attached how many buttons are present or not.

---------- Cut here ----------

4. (a) Write a VC++ program to handle windows messages in MFC program.

(b) Create a worksheet with the following field student Name, Internal mark, External mark and Result. Using data access object. Create a student application to do the following :

(i) Insertion

(ii) Deletion

(iii) Updation.

5. (a) Write a VC++ program to get the status of the shift and toggle keys using MFC.

   (b) Using MS Flex grid control display the multiplication and addition table of 20 rows and columns.

---------- Cut here ----------

6. (a) Write an event procedure for the following:

   (i) Display the date and time in label box during run time.

   (ii) Find the Power of a number.

   (iii) Temperature conversion.

   (Use Label, Text box, Command Button)

   (b) Write a VC++ program to create window of desired size using MFC.

---------- Cut here ----------

7. (a) Using built in AX control, to develop the windows NOTEPAD with file and edit menu operation and also display the floating menu whenever necessary.

   (b) Write a VC++ program to create a list box in window.

---------- Cut here ----------

8. (a) Write a VC++ program to generate a status box and show the status of Caps Lock, Num Lock and Scroll Lock in it.

   (b) Write an event procedure for the following :

   (i) Reverse a string

   (ii) Determine whether the given string is palindrome or not.

DE–9217

Sub. Code
47

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

WEB TECHNOLOGY – LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Create a web page in the format of front page of a e-commerce Rowspan and colspan with hyper link.

   (b) Write Servlet program to send Email message to a user.

---------- Cut here ----------

2. (a) Write an XML document to display your Business Card. Write an XSL style sheet and attach that to the XML document. Validate the document using DTD or XSD.

   (b) Create a web page in the format of front page of a College using Text links also align the text with colors.

3. (a) Write a Servlet program to display "Your Name".

   (b) Develop an Application for JSP – Servlet communication.

---------- Cut here ----------

4. (a) Write an XML document to display your bio-data. Write an XSL style sheet and Attach that to the XML document. Validate the document using DTD or XSD.

   (b) Create a HTML web page in the format of front page of Book store with Book advertisement animation using Java Script.

---------- Cut here ----------

5. (a) Write a JSP program to display student mark list using JDBC.

   (b) Write Servlet program to send Email message.

---------- Cut here ----------

6. (a) Write a JSP program to display student mark list using JDBC.

   (b) Create a Image animation using Java script.

7. (a) Write an ASP program to prepare student performance evaluation document using Java script.

(b) Create a Student mark list using HTML table with rows and columns and split them using Rowspan and Colspan.

---------- Cut here ----------

8. (a) Write a JSP program to display student Attendance list for 12 months using JDBC.

(b) Create a HTML table with rows and columns for Cricket Score card with player information such as name, Score.

DE–3924

Sub. Code
51

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

SOFTWARE TESTING AND REUSE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the purpose of testing.
2. Discuss the consequences of bugs.
3. How do you implement transaction flow testing?
4. Discuss on nice and ugly domain.
5. Write a note on Decision table.
6. What are state graphs? Explain.
7. Discuss the powers of a matrix.
8. Explain about rapid testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss on different taxonomy of bugs and model for testing.
10. Explain the basic concepts of path testing.

11. Explain Data flow model.

12. Discuss on domains and interface testing.

13. Describe about Regular expressions and flow anomaly detection.

14. Explain Node reduction Algorithm.

15. Explain about performance testing of a data base application and HTTP connection for website access.

DE–3925

Sub. Code
52

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. List the different types of Security Attacks.

2. Explain Internet standards.

3. Write short notes on message authentication without message encryption.

4. Explain principles of Public key cryptography.

5. Write short notes on digital signatures.

6. Explain web security requirements.

7. Discuss Encapsulating Security Payload.

8. Explain the effects of viruses in detail.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe the security services in detail.

10. Explain key distribution approaches.

11. Discuss Secure Hash Functions.

12. Explain digital signatures.

13. Write short notes on IP security overview.

14. Describe Secure Electronic Transaction.

15. Explain Intrusion Detection System in detail.

DE–3926

Sub. Code
53

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

SOFTWARE QUALITY ASSURANCE AND STANDARDS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Explain the metrics needed to measure the quality of a software.

2. Write a note on: Gilb's Approach

3. Discuss the steps that are carried out in reviews and audits to assess the software quality.

4. Explain the tools involved in determining the quality of a software.

5. Enlist the steps to prevent the defects and to remove them.

6. Give a brief account on the elements of Quality Management System.

7. Briefly explain the need of standards to predict the quality of a software.

8. Explain the characters that are influenced to assure the quality of a software.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe in detail Hierarchical models of Boehm and McCall.

10. Elucidate the steps involved in Software Quality Assurance Plan.

11. Discuss about reliability growth model for quality assessment.

12. Comment on complexity metrics and models.

13. Explain the standards that are incurred in ISO 9000 series to upgrade the quality of a software.

14. Explain the CASE tools that are used to justify the quality of a software.

15. Discuss the steps to be adopted before implementing a software.

Ws1

DE–3927

Sub. Code
54

DISTANCE EDUCATION

M.Sc (SOFTWARE ENGINEERING)
(5 YEARS INTEGRATED). DEGREE EXAMINATION,
MAY 2016.

MOBILE COMMUNICATIONS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write about system architecture of GSM.

2. Explain about the architecture of Bluetooth.

3. Explain about dynamic host configuration protocol.

4. Explain about indirect TCP.

5. Explain about the architecture of wireless application protocol.

6. Write short notes on.

   (a) Fixed TDMA

   (b) Classical Aloha

   (c) Slotted Aloha.

7. Write about MAC management in IEEE 802.11.

8. Explain about digital audio broadcasting.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain about routing, localization, handover in satellite systems.

10. Explain about system, protocol, architecture of IEEE 802.11.

11. Explain about routing and dynamic source routing in mobile ad–hoc networks.

12. Explain about transaction oriented TCP.

13. Discus on wireless telephony application.

14. Explain about 802.11b and 802.11a in IEEE 802.11.

15. Discuss on the history of HIPHERLAN.

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DE–3928

Sub. Code
55

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2016.

OPEN SOURCE ARCHITECTURE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss on history and advantages of open source software.

2. Explain the steps for the OS boot process.

3. Explain how to setup firewall using the GNU compiler.

4. Explain how to use autoconf and autogen files for different environment.

5. Explain the linking against object archivers and dynamic shared object libraries

6. Explain GTK programming with example program to print “hello world”.

7. Explain about system configuration files and file logs in FOSS.

8. Explain in detail how to setup proxy server.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain how to partition the disk and file systems and installation of LINUX.

10. Explain how to perform every day task using GNU/LINUX.

11. Explain in detail about C compiler and C++ compiler.

12. Explain about static and dynamic linked libraries.

13. Discuss about basics of TCP/IP networking and routing.

14. Explain how to setup printer services and file services.

15. Discuss the basics of X windows server architecture.

DE–9218

Sub. Code
56

DISTANCE EDUCATION

M.Sc. (SE) DEGREE EXAMINATION, MAY 2016.

Software Engineering (5 Year Integrated)

OPEN SOURCE LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a PHP program interface to create a database and to insert a table into it.

   (b) Write a PHP program to create, insert, update, display, and delete student information as follows Name, Roll no, Marks, and Grade.

---------- Cut here ----------

2. (a) Write a PHP program that adds products that are selected from a web page to a shopping cart.

   (b) Write a shell program to find the details of a user session.

---------- Cut here ----------

3. (a) Write a PHP program to access the data from my sql table.

   (b) Write a PHP program using classes to create a table.

4. (a) Write a PHP program to upload a file to the server.

   (b) Create my sql table and execute queries to read, add, remove and modify a record from that table.

---------- Cut here ----------

5. (a) Write a PHP program to create a director, and to read contents from the directory.

   (b) To develop a PHP program for display the selected image using the combo box control.

---------- Cut here ----------

6. (a) Create a program in PHP to display the student marks which are greater than 80.

   (b) Write a code in PHP script to create the Registration form for inter-college competition.

---------- Cut here ----------

7. (a) PHP code to create a simple web site.

   (b) Write a PHP program to read and write a text in a file.

---------- Cut here ----------

8. (a) Write a PHP program to perform string comparison, string concatenation, and string length.

   (b) To develop a PHP program for open, read and close the file.

DE–9219

Sub. Code
57

DISTANCE EDUCATION

M.Sc. (SE) DEGREE EXAMINATION, MAY 2016.

Software Engineering (5 Year Integrated)

CASE TOOLS — LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Perform program analysis and project planning to identify project scope, objective and infrastructure.

   (b) Develop a stock maintenance system and perform test case hierarchy site check and site monitor.

---------- Cut here ----------

2. (a) Develop a Software to illustrate the method of debugging.

   (b) Develop a real-time schedule and perform overage analysis and memory leaks check.

---------- Cut here ----------

3. (a) Perform software requirement analysis that describes modules of the project and identifies deliverables.

   (b) Develop an expert system and perform coverage analysis and memory leaks check.

4. (a) Implement an ATM system and perform validation testing.

   (b) Develop a course registration System and perform coverage analysis and memory leaks check.

---------- Cut here ----------

5. (a) Develop an online ticket reservation system and perform validation testing.

   (b) Perform software requirement analysis to describe the modules of a project and identity deliverables.

---------- Cut here ----------

6. (a) Develop a student analysis to describe the modules of a project and to identify deliverables.

   (b) Illustrate the method of generating data dictionary using a suitable software.

---------- Cut here ----------

7. (a) Develop a pay roll system and perform validation testing.

   (b) Develop a quiz system and perform coverage analysis.

---------- Cut here ----------

8. (a) Develop a software to illustrate the concept of debugging.

   (b) Illustrate data modelling through use diagrams and activity diagram.